

Circuit Documentation

Summary

This circuit is a complex system integrating various sensors, a motor driver, a servo motor, and a Raspberry Pi 3B+ as the central processing unit. The circuit is designed to control a motor and a servo, read data from multiple sensors, and manage power distribution efficiently. The Raspberry Pi interfaces with the components through GPIO pins, and the system is powered by a 12V battery with a step-down converter to provide necessary voltage levels for different components.

Component List

1. **TB6612FNG Motor Driver**
 - Description: A dual motor driver capable of controlling two DC motors with PWM input.
 - Pins: GND, B01, B02, A02, A01, VCC, VM, PWMB, BI2, BI1, STBY, AI1, AI2, PWMA
2. **SG90 Servo Motor**
 - Description: A small servo motor used for precise control of angular position.
 - Pins: PWM, +5V, GND
3. **ToF400C-VL53L1X (x4)**
 - Description: Time-of-Flight distance sensor for measuring distances.
 - Pins: SHUT, INT, SCL, SDA, GND, VIN
4. **XL4015 5A DC Buck Step-down**
 - Description: A step-down voltage converter to provide stable voltage output.
 - Pins: Output +, Output -, Input +, Input -
5. **TCS34725**
 - Description: A color sensor with an integrated IR filter and white LED.
 - Pins: VIN, GND, 3.3V, SCL, SDA, INT, LED
6. **10 DOF IMU Sensor**
 - Description: An inertial measurement unit sensor for motion tracking.
 - Pins: VCC, GND, SDA, SCL, INT, FSYNC
7. **Pushbutton**
 - Description: A simple pushbutton switch for user input.
 - Pins: Pin 2, Pin 1, Pin 3, Pin 4
8. **Switch**
 - Description: A basic on/off switch for controlling power.
 - Pins: Vin, GND, Vout
9. **Resistor**
 - Description: A 10k Ohm resistor used for current limiting or pull-up/pull-down.
 - Pins: pin1, pin2
10. **Ceramic Capacitor (x2)**
 - Description: A 0.001 Farad capacitor for filtering and stabilization.
 - Pins: pin0, pin1

11. Logic Level Shifter Converter Module

- Description: A module for shifting logic levels between different voltage domains.
- Pins: HV1, HV2, HV, GND, HV3, HV4, LV1, LV2, LV, LV3, LV4

12. Raspberry Pi 3B+

- Description: A single-board computer used as the main controller.
- Pins: Multiple GPIO pins, power pins, and communication interfaces.

13. Encoder Metal Gearmotor 12V DC High Speed 300RPM

- Description: A high-speed DC motor with an integrated encoder for feedback.
- Pins: Quad encoder B signal, Quad encoder A signal, Quad encoder +5V Vcc, Quad encoder Ground, Motor power terminal(-), Motor power terminal (+)

14. Battery 12V

- Description: A 12V battery used as the primary power source.
- Pins: +, -

Wiring Details

TB6612FNG Motor Driver

- **GND**: Connected to the common ground.
- **B01**: Connected to the Motor power terminal (+) of the Encoder Metal Gearmotor.
- **B02**: Connected to the Motor power terminal (-) of the Encoder Metal Gearmotor.
- **STBY**: Connected to GPIO 23 on the Raspberry Pi.
- **BI1**: Connected to GPIO 5 on the Raspberry Pi.
- **BI2**: Connected to GPIO 6 on the Raspberry Pi.
- **PWMB**: Connected to HV2 on the Logic Level Shifter Converter Module.
- **VM**: Connected to Vout on the Switch.
- **VCC**: Connected to the 5V Power from the Raspberry Pi.

SG90 Servo Motor

- **PWM**: Connected to HV1 on the Logic Level Shifter Converter Module.
- **+5V**: Connected to the 5V Power from the Raspberry Pi.
- **GND**: Connected to the common ground.

ToF400C-VL53L1X Sensors

- **VIN**: Connected to the 3.3V Power from the Raspberry Pi.
- **GND**: Connected to the common ground.
- **SDA**: Connected to GPIO 2 (I2C1 SDA) on the Raspberry Pi.
- **SCL**: Connected to GPIO 3 (I2C1 SCL) on the Raspberry Pi.
- **SHUT**: Connected to various GPIO pins on the Raspberry Pi for each sensor.

XL4015 5A DC Buck Step-down

- **Input +**: Connected to Vin on the Switch.

- **Input -**: Connected to GND on the Switch.
- **Output +**: Connected to the 5V Power from the Raspberry Pi.
- **Output -**: Connected to the common ground.

TCS34725

- **VIN**: Connected to the 3.3V Power from the Raspberry Pi.
- **GND**: Connected to the common ground.
- **SDA**: Connected to GPIO 2 (I2C1 SDA) on the Raspberry Pi.
- **SCL**: Connected to GPIO 3 (I2C1 SCL) on the Raspberry Pi.

10 DOF IMU Sensor

- **VCC**: Connected to Pin 1 of the Pushbutton.
- **GND**: Connected to the common ground.
- **SDA**: Connected to GPIO 2 (I2C1 SDA) on the Raspberry Pi.
- **SCL**: Connected to GPIO 3 (I2C1 SCL) on the Raspberry Pi.

Pushbutton

- **Pin 1**: Connected to VCC of the 10 DOF IMU Sensor.
- **Pin 3**: Connected to pin1 of the Resistor and GPIO 17 on the Raspberry Pi.

Switch

- **Vin**: Connected to the positive terminal of the Battery 12V.
- **GND**: Connected to the negative terminal of the Battery 12V and common ground.
- **Vout**: Connected to VM on the TB6612FNG Motor Driver.

Resistor

- **pin1**: Connected to Pin 3 of the Pushbutton.
- **pin2**: Connected to GND on the Raspberry Pi.

Ceramic Capacitors

- **pin0**: Connected to the 5V Power from the Raspberry Pi.
- **pin1**: Connected to the common ground.

Logic Level Shifter Converter Module

- **HV**: Connected to the 5V Power from the Raspberry Pi.
- **GND**: Connected to the common ground.
- **LV**: Connected to the 3.3V Power from the Raspberry Pi.
- **LV1**: Connected to GPIO 18 (PCM CLK) on the Raspberry Pi.
- **LV2**: Connected to GPIO 13 (PWM1) on the Raspberry Pi.

Raspberry Pi 3B+

- **GPIO 2 (I2C1 SDA):** Connected to SDA of the sensors.
- **GPIO 3 (I2C1 SCL):** Connected to SCL of the sensors.
- **GPIO 5:** Connected to BI1 on the TB6612FNG Motor Driver.
- **GPIO 6:** Connected to BI2 on the TB6612FNG Motor Driver.
- **GPIO 13 (PWM1):** Connected to LV2 on the Logic Level Shifter Converter Module.
- **GPIO 17:** Connected to Pin 3 of the Pushbutton.
- **GPIO 18 (PCM CLK):** Connected to LV1 on the Logic Level Shifter Converter Module.
- **GPIO 23:** Connected to STBY on the TB6612FNG Motor Driver.
- **GPIO 24:** Connected to Quad encoder B signal of the Encoder Metal Gearmotor.
- **GPIO 25:** Connected to SHUT of one ToF400C-VL53L1X sensor.
- **GPIO 26:** Connected to SHUT of another ToF400C-VL53L1X sensor.
- **5V Power:** Connected to VCC of the TB6612FNG Motor Driver and +5V of the SG90 Servo Motor.
- **3.3V Power:** Connected to VIN of the ToF400C-VL53L1X sensors and TCS34725.

Encoder Metal Gearmotor 12V DC High Speed 300RPM

- **Quad encoder +5V Vcc:** Connected to Pin 1 of the Pushbutton.
- **Quad encoder Ground:** Connected to the common ground.
- **Motor power terminal (+):** Connected to B01 on the TB6612FNG Motor Driver.
- **Motor power terminal (-):** Connected to B02 on the TB6612FNG Motor Driver.

Battery 12V

- **+: Connected to Vin on the Switch.**
- **-: Connected to GND on the Switch.**

Documented Code

```
import RPi.GPIO as GPIO
import time

# Pins
AIN1 = 17
AIN2 = 27
PWMA = 22
STBY = 23

GPIO.setmode(GPIO.BCM)
GPIO.setup(AIN1, GPIO.OUT)
GPIO.setup(AIN2, GPIO.OUT)
GPIO.setup(PWMA, GPIO.OUT)
GPIO.setup(STBY, GPIO.OUT)
```

```
# Enable driver
GPIO.output(STBY, GPIO.HIGH)

# PWM setup
pwm = GPIO.PWM(PWMA, 1000) # 1kHz PWM
pwm.start(0)

# Forward at 70% speed
GPIO.output(AIN1, GPIO.HIGH)
GPIO.output(AIN2, GPIO.LOW)
pwm.ChangeDutyCycle(70)
time.sleep(5)

# Reverse at 50% speed
GPIO.output(AIN1, GPIO.LOW)
GPIO.output(AIN2, GPIO.HIGH)
pwm.ChangeDutyCycle(50)
time.sleep(5)

# Stop motor
GPIO.output(AIN1, GPIO.LOW)
GPIO.output(AIN2, GPIO.LOW)
pwm.ChangeDutyCycle(0)
GPIO.output(STBY, GPIO.LOW)

GPIO.cleanup()
```

This code is designed to control a motor using the TB6612FNG Motor Driver connected to a Raspberry Pi. It sets up the GPIO pins, enables the motor driver, and uses PWM to control the motor speed and direction. The motor runs forward at 70% speed, then reverses at 50% speed, and finally stops.